

Typical Stresses and Their Effects Table 3

Typical stresses	Effects
High temperature	Dielectric breakdown, electrical parameter shift
Low temperature	Water condensation, electrical parameter drift
Thermal cycling	PWA opens, shorts and seal failures
Vibration	Wire chafing, loosening of hardware
Humidity	Corrosion of materials, breakdown of insulations
Low pressure	Reduction of electrical breakdown voltages

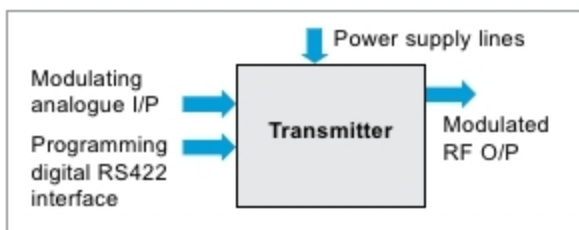


Fig. 1: Typical I/Os for a transmitter unit

to test the unit. Some connector pins carry inputs to unit, while some carry outputs. Some are bi-directional, while others may carry power to other units. Typical I/Os for a transmitter unit are given in Fig. 1.

4. Application stress levels. Every product encounters a number of stresses during operation, storage, handling and transportation. Examples of stresses are temperature (high and low), thermal shock, humidity, mechanical shock, vibration and so on. These stresses can cause failure of solder joints, mechanical joints and components by thermal, mechanical and chemical actions. Hence, the test engineer needs to know all these stresses and their values applicable to the product. Typical stresses and their effects on product performance are given in Table 3, and this list is not exhaustive.

5. Regulatory requirements. Local regulatory/standards body requirements of the product must also be adhered to; for example, Pollution Control Body and electromagnetic interference (EMI)/electromagnetic compatibility (EMC). EMI/EMC requirements are important both from regulatory and application points of view.

6. Functional test plan. A functional test plan at ambient conditions has to be arrived at after obtaining all required information. Products are usually tested by applying inputs of appropriate values that occur during application. Analogue inputs can be tested for their entire dynamic range as per design by applying input over the entire range or as per application.

Testing over the entire dynamic range assures good performance of product, even if application requirements change. Expected outputs for inputs should be measured. Digital inputs should be tested by applying levels, patterns and bit rates as per application and logic family/standard of inputs.

RF/microwave receivers should be tested by applying input signal of the appropriate frequency and level with modulation. RF/microwave transmitters must be tested by applying appropriate modulation input. Care should be taken to ensure all I/Os and performance specifications are covered in the test plan.

Some outputs may be DC voltage lines to drive power to other sub-systems. These should not be missed during testing, since their failure can appear later after integration with other sub-systems.

Every product must perform well over a range of supply voltage variations, which are likely to occur during operation and in the installation facility. The test plan should have test cases to ensure that the product performs well over the allowed supply voltage range.

7. Testing for regulatory requirements. After ensuring a functional test plan that the product meets all performance and I/O specifications, alongwith application requirements, the product should be tested to ensure that it meets EMI/EMC and other regulatory requirements. EMI/EMC requirements are important to assure that the product will perform well in the presence of EMI from other electronic systems and that this would not affect performance due to its own electromagnetic emissions.

8. Testing for stress requirements. As indicated earlier, the test plan should have test cases to ensure that the product performs well and meets its functionality and specification requirements under identified stresses that occur during operation, handling and storage. Exhaustive testing may not be possible here. Critical parameters must be identified and tested for compliance before, during

and after application of stresses for the required duration.

For transportation stresses, operation before and after application of stress should be ensured. Certified environmental test labs and calibrated test instruments should be utilised for this purpose.

9. Testing before delivery. Exhaustive testing as per the functional test plan has to be repeated after completion of stress compliance tests before delivery of the unit. This is because only critical parameters are covered in stress tests, and any other minor failures during stresses will not be known unless exhausting testing is repeated.

10. Documentation. All information collected in the above steps must be converted into a test plan document. This document should have top-level diagrams of the unit showing power and I/Os, description of functionality and working of the unit with a supporting logical block diagram, specifications of functionality, performance, environmental stresses, and regulatory and EMI/EMC requirements.

It should also have a detailed functional test procedure describing how to measure all performance specifications, along with the test setup. Care should be taken to ensure all performance specs and I/Os are covered in the test plan. It should have test results recording formats or test leaves with acceptance criteria to record measured test results in ambient conditions and under various environmental stress conditions.

Finally, it should have a section called conformance report for recording unit information (serial number, model number, project number, etc) and sequence of the tests conducted, any observations/issues in the tests conducted and clearance status of the unit for the intended application.

Conclusion

Tests done as per the document that is prepared and approved as per information under section 10 and test leaves filled as per that document and conformance report for the unit will provide confidence to the management and the customer that the product meets all requirements for the intended application. This is when an electronic quality test engineer's job is accomplished. **EFY**